Exp3: Build a Docker container to run a Python script that performs basic file operations

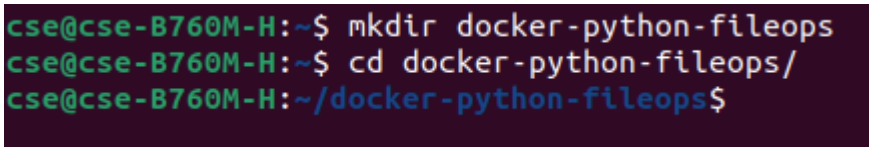
Software Requirements

- Ubuntu 22.04 LTS
- Docker Engine
- Python (inside container)

Step 1: Create Project Directory

```
mkdir docker-python-fileops
```

```
cd docker-python-fileops
```



Step 2: Create Python Script (File Operations)

Create a file named **file_ops.py**:

```
nano file_ops.py
```

Paste the following code:

```
# Create a file and write data
with open("sample.txt", "w") as file:
    file.write("This file is created inside a Docker container.\n")

# Append data
with open("sample.txt", "a") as file:
    file.write("File operations using Python in Docker.\n")

# Read and display file content
with open("sample.txt", "r") as file:
    content = file.read()
```

```
print("Contents of the file:")

print(content)
```

```
  GNU nano 6.2
# Create a file and write data
with open("sample.txt", "w") as file:
    file.write("This file is created inside a Docker container.\n")

# Append data
with open("sample.txt", "a") as file:
    file.write("File operations using Python in Docker.\n")

# Read and display file content
with open("sample.txt", "r") as file:
    content = file.read()

print("Contents of the file:")
print(content)
```

Save and exit.

Step 3: Create Dockerfile

Create **Dockerfile**:

```
nano Dockerfile
```

Paste:

```
FROM python:3.10-slim

WORKDIR /app

COPY file_ops.py .

CMD ["python", "file_ops.py"]
```

```
  GNU nano 6.2
FROM python:3.10-slim

WORKDIR /app

COPY file_ops.py .

CMD ["python", "file_ops.py"]
```

Save and exit.

Step 4: Build Docker Image

```
docker build -t python-fileops .
```

```
cse@cse-B760M-H:~/docker-python-fileops$ docker build -t python-fileops .
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
            Install the buildx component to build images with BuildKit:
            https://docs.docker.com/go/buildx/

Sending build context to Docker daemon  3.072kB
Step 1/4 : FROM python:3.10-slim
 ---> 988b1bc80158
Step 2/4 : WORKDIR /app
 ---> Using cache
 ---> 7577978671de
Step 3/4 : COPY file_ops.py .
 ---> 309f43ec1b5b
Step 4/4 : CMD ["python", "file_ops.py"]
 ---> Running in 8e46c883f84e
 ---> Removed intermediate container 8e46c883f84e
 ---> fe055b321a21
Successfully built fe055b321a21
Successfully tagged python-fileops:latest
cse@cse-B760M-H:~/docker-python-fileops$
```

```
docker images
```

```
cse@cse-B760M-H:~/docker-python-fileops$ docker images
REPOSITORY          TAG         IMAGE ID       CREATED              SIZE
python-fileops      latest      fe055b321a21   About a minute ago   122MB
simple-web-app      latest      5557786ca37d   2 days ago           134MB
hello-world-image   latest      b7e187b4b9ab   2 days ago           77.9MB
python              3.10-slim   988b1bc80158   11 days ago          122MB
ubuntu              22.04       65c77cbc27c2   5 weeks ago          77.9MB
cse@cse-B760M-H:~/docker-python-fileops$
```

Step 5: Run the Docker Container

```
docker run python-fileops
```

Expected Output:

Contents of the file:

This file is created inside a Docker container.

File operations using Python in Docker.

```
cse@cse-B760M-H:~/docker-python-fileops$ docker run python-fileops
Contents of the file:
This file is created inside a Docker container.
File operations using Python in Docker.

cse@cse-B760M-H:~/docker-python-fileops$
```

```
docker ps -a
```

```
cse@cse-B760M-H:~/docker-python-fileops$ docker ps -a
CONTAINER ID   IMAGE              COMMAND                  CREATED         STATUS                      PORTS     NAMES
38274fb5d628   python-fileops     "python file_ops.py"     2 minutes ago   Exited (0) 2 minutes ago              frosty_mayer
b116a6238394   simple-web-app     "python app.py"          2 days ago      Exited (137) 2 days ago               competent_cray
72326bcd11bc   hello-world-image  "echo 'Hello World'"     2 days ago      Exited (0) 2 days ago                 eloquent_dirac
cse@cse-B760M-H:~/docker-python-fileops$
```

Step 6: Verify File Inside Container

Run an interactive container:

```
docker run -it python-fileops /bin/bash
```

Inside container:

```
ls
```

```
cat sample.txt
```

```
cse@cse-B760M-H:~/docker-python-fileops$ docker run -it python-fileops /bin/bash
root@daefddb6f37b:/app# ls
file_ops.py
root@daefddb6f37b:/app# cat sample.txt
cat: sample.txt: No such file or directory
root@daefddb6f37b:/app#
```

Important Notes:

Note 1:

What we did?

We ran the container normally:

`docker run python-fileops`

This executed `file_ops.py`, created `sample.txt`, printed output, then exited.

*Then we ran **a new container** interactively:*

`docker run -it python-fileops /bin/bash`

Inside this new container:

`ls`

`cat sample.txt`

Result is: sample.txt not found

Because each `docker run` command creates a NEW container. Containers do not share files with each other by default.

Why sample.txt is missing?

First container:

- *Ran file_ops.py*
- *Created sample.txt*
- *Printed content*
- *Container **stopped***
- *File exists **only inside that container***

Second container:

- *Brand new container*
- *Same image, **fresh filesystem***
- *file_ops.py was **NOT executed***
- *So sample.txt was **never created***

Docker images are read-only templates.

Docker containers are runtime instances.

Files created at runtime stay inside that specific container only.

Let us prove this now:

Run interactive container

```
docker run -it python-fileops /bin/bash
```

Run the script manually

```
python file_ops.py
```

Check file

```
ls -la
cat sample.txt
```

Now the file WILL exist.

```
exit
```

```
cse@cse-B760M-H:~/docker-python-fileops$ docker run -it python-fileops /bin/bash
root@daefddb6f37b:/app# ls
file_ops.py
root@daefddb6f37b:/app# cat sample.txt
cat: sample.txt: No such file or directory
root@daefddb6f37b:/app#
root@daefddb6f37b:/app#
root@daefddb6f37b:/app# python file_ops.py
Contents of the file:
This file is created inside a Docker container.
File operations using Python in Docker.

root@daefddb6f37b:/app#
root@daefddb6f37b:/app#
root@daefddb6f37b:/app# ls -la
total 16
drwxr-xr-x 1 root root 4096 Feb 16 06:07 .
drwxr-xr-x 1 root root 4096 Feb 16 05:41 ..
-rw-rw-r-- 1 root root  394 Feb 16 05:34 file_ops.py
-rw-r--r-- 1 root root   88 Feb 16 06:07 sample.txt
root@daefddb6f37b:/app#
root@daefddb6f37b:/app#
root@daefddb6f37b:/app# cat sample.txt
This file is created inside a Docker container.
File operations using Python in Docker.
root@daefddb6f37b:/app# exit
exit
cse@cse-B760M-H:~/docker-python-fileops$
```

Note 2:

Files created inside a container exist only inside that container unless volumes are used.

Once the container stops:

- *File still exists* ***inside the container***
- *File is* ***NOT visible on host system***

(This leads naturally to Docker volumes in later labs.)

Result:

A Docker container was successfully built and executed to perform basic file operations using a Python script.